

TECHBIG SOLUTIONS

Ambattur, Chennai

SAP ATTP – Complete E2E Training Syllabus

SAP Advanced Track and Trace for Pharmaceuticals

Program objective: Develop practical consultant-level skills in pharmaceutical serialization, aggregation, tracking and tracing, regulatory reporting, ECC/ERP integration, warehouse and packaging-line integration, configuration, testing and production support.

Reference basis: SAP Help documentation for SAP Advanced Track and Trace for Pharmaceuticals (ATTP), including the ECC Add-On/Serialization Connector, serialization profiles, serial-number management, event processing, warehouse integration and India-specific reporting.

Program	SAP ATTP – Complete E2E
Level	Functional Consultant / Implementation / Integration / Support
Core Areas	Serialization Aggregation EPCIS Repository Regulatory Reporting
Integration	SAP ECC/ERP WM/EWM Packaging Lines 3PL/CMO External Authorities
Suggested Duration	80–120 hours, customizable by project and batch requirements

1. SAP ATTP Overview & Pharmaceutical Track-and-Trace

- Purpose and business value of SAP Advanced Track and Trace for Pharmaceuticals (ATTP)
- Serialization, aggregation, tracking, tracing and authentication
- Country-specific regulatory compliance and reporting
- ATTP repository concept and central serial-number event management
- Integration with ERP/ECC, warehouse systems, packaging lines, 3PLs and CMOs

2. ATTP Architecture & System Landscape

- ATTP application/repository architecture
- SAP NetWeaver and Application Interface Framework foundation
- ECC Add-On / Serialization Connector architecture
- Development, quality and production landscape
- Inbound/outbound communication concepts
- Systems, business partners and external applications

3. Pharmaceutical Master Data

- Trade items and GTIN concepts
- Locations, plants and storage locations
- Customers, vendors and business partners
- Batches and batch relationships
- Packaging levels and hierarchy
- Serial-number relevant master-data attributes
- Country-specific product data

4. Serialization Fundamentals

- Serialization types and serialized trade items
- SGTIN, SSCC and LGTIN concepts
- Serial-number lifecycle

- Internal vs external serial-number management
- Uniqueness scope
- Serial-number status management
- Serialization compliance requirements

5. Serialization Profiles & Number Ranges

- Serialization profiles
- Range definitions and ranges
- Range requests and range consumers
- Range automation profiles
- Sequential and randomized serial numbers
- Country/business-partner assignment
- Number-range activation and closing
- Serial-number generation and validation

6. Trade Item Serialization Configuration

- Activate trade items for serialization
- Assign serialization profiles
- Assign range definitions
- Configure serialization attributes
- Product category and country relevance
- Internal/external number management
- Troubleshooting activation and master-data errors

7. Aggregation & Packaging Hierarchies

- Primary, secondary, intermediate and tertiary packaging
- Aggregation and disaggregation
- Parent-child relationships
- Commissioning and decommissioning
- Packing/unpacking events
- Hierarchy validation and accuracy
- Aggregation reconciliation

8. EPCIS & Event Management

- EPCIS fundamentals
- Object, aggregation, transaction and transformation events
- Event data and business context
- EPCIS message structure
- Inbound event processing
- Event validation and object updates
- Late-event processing and event relationships

9. ATTP Repository & Object Management

- Repository objects and object lifecycle
- Serialized trade-item objects

- Serialized containers
- Object hierarchy and relationships
- Business transactions and event relations
- Object status management
- Repository search and operational analysis

10. ECC Integration – Master Data

- ECC Add-On / Serialization Connector
- Material master integration
- Customer and vendor integration as business partners
- Plant and storage-location integration
- Batch integration
- Trade-item master-data enrichment
- Integration monitoring

11. ECC Integration – Transactional Data

- Production/process order integration
- Inbound, outbound and return deliveries
- Batch-related transactions
- Goods movements and serialization events
- Document relationships
- Transaction reconciliation between ECC and ATTP

12. Warehouse Integration

- SAP WM / warehouse-process integration
- EWM integration concepts
- Serialization validation during warehouse scanning
- Inbound receiving and putaway
- Picking, packing, loading and shipping
- Container receive/ship
- Online validation and buffering
- EPCIS event composition after warehouse process completion

13. Packaging Line Integration

- Packaging-line serialization concepts
- Serial-number range/list requests
- Commissioning from packaging lines
- Aggregation capture
- Decommissioning/rework concepts
- Interface monitoring
- Line integration testing and reconciliation

14. Business Partner / CMO Integration

- Marketing Authorization Holder (MAH) and CMO scenarios
- External serial-number management

- Serial-number requests
- Range/list distribution
- Supply-chain partner communication
- Inbound/outbound serialization events
- Partner reconciliation

15. Regulatory Reporting

- Regulatory reporting architecture
- Reporting rules and reporting events
- Country-specific reporting requirements
- Legal reporting monitoring
- XML/message generation concepts
- Submission status and error handling
- Audit trail and evidence management

16. India ATTP / DAVA Reporting

- India pharmaceutical serialization requirements
- DAVA portal concepts
- Export-drug serialization and legal reporting
- Indian packaging hierarchy
- Production details reporting
- Batch details reporting
- India-specific reporting troubleshooting

17. Country-Specific Compliance

- Country-specific serialization rules
- Number format and randomization requirements
- Data-capture and reporting requirements
- Country profiles and system configuration
- Examples: India, China, Brazil and other regulated markets
- Regulatory change management

18. Verification & Authentication

- Batch authentication concepts
- Serialized trade-item verification
- Serial-number status validation
- Verification interfaces
- Suspect/counterfeit handling concepts
- Response processing and auditability

19. Event Processing & Rules

- Event decoding and validation
- BAdI-based enhancement concepts
- Auto-create and auto-unpack
- Late-event processing

- Event-object relationships
- Business transactions
- Rule processing and automated actions

20. Application Interface Framework (AIF)

- AIF architecture and purpose
- Interface configuration
- Monitoring failed messages
- Error handling and reprocessing
- Mapping and value mapping concepts
- Custom enhancements
- Production monitoring and support

21. Communication Interfaces

- OData service concepts
- RFC communication
- XML and message interfaces
- EPCIS communication
- Interface security and technical users
- Synchronous vs asynchronous processing
- Connectivity troubleshooting

22. ATTP Warehouse Toolbox

- Warehouse business functions
- Serialization check
- Validate activity and buffer
- Compose/send EPCIS event
- Inbound, internal and outbound activities
- Pack/unpack and aggregation
- Inspect hierarchy and aggregation accuracy
- RF and custom warehouse-process integration

23. ATTP Monitoring & Reporting

- Repository monitoring
- Serial-number usage reports
- Event monitoring
- Business transaction monitoring
- Regulatory reporting monitoring
- Interface/AIF monitoring
- Data consistency and reconciliation

24. Security & Authorizations

- ATTP user administration
- Roles and authorization concepts
- Technical users and communication security

- Business-partner/system access
- Segregation of duties
- Audit and production-access controls

25. Configuration / IMG & Customizing

- ATTP repository customizing
- Serialization settings
- Range and serialization configuration
- Trade-item configuration
- Rule customizing
- System and partner configuration
- Interface configuration
- Transport and change management

26. Implementation Methodology

- Requirement gathering and regulatory assessment
- Process mapping
- Master-data preparation
- Serialization design
- Integration design
- Configuration and development
- Unit/SIT/UAT testing
- Cutover and go-live
- Hypercare and stabilization

27. Testing & Validation

- Serialization test cases
- Commissioning and aggregation tests
- Decommission/rework tests
- Warehouse integration tests
- ECC document-flow tests
- EPCIS message validation
- Regulatory-reporting tests
- Negative/error test scenarios
- Reconciliation and audit testing

28. Production Support & Troubleshooting

- Missing serial numbers
- Invalid range/range-request errors
- Hierarchy inconsistencies
- EPCIS/event-processing errors
- ECC-ATTP data mismatches
- Warehouse validation failures
- AIF/interface errors

- Regulatory reporting failures
- Root-cause analysis and incident management

29. Real-Time E2E Project Scenarios

- Pharma product master → serialization profile → serial-number range → commissioning
- Packaging line → aggregation → EPCIS events → ATTP repository
- ECC production order → serialization → packing hierarchy → finished-goods release
- ECC outbound delivery → warehouse pick/pack → ship event → regulatory reporting
- CMO receives external serial-number range → production → commissioning → reporting
- India export product → multi-level aggregation → DAVA-related reporting
- 3PL warehouse integration with serialized inbound/outbound processing
- Serial-number exception → investigation → correction → event reconciliation
- Product recall/traceability investigation using serialized hierarchy
- Complete ATTP implementation and production-support project

30. Interview & Practical Assessment

- ATTP consultant interview questions
- Serialization and aggregation scenarios
- ECC integration and CIF-style interface troubleshooting concepts
- EPCIS/event-processing questions
- Regulatory-reporting case studies
- Configuration case study
- Production-support ticket simulation
- End-to-end project walkthrough and mock interview

Recommended Hands-on Deliverables

- Configure a serialized trade item with serialization profile, range definition and number management.
- Execute commissioning, aggregation, disaggregation and decommissioning scenarios.
- Integrate ECC master and transactional data with ATTP and reconcile the document flow.
- Configure and test warehouse serialization validation and EPCIS event generation.
- Build and troubleshoot regulatory reporting flows, including India-focused scenarios.
- Analyze repository objects, event histories, hierarchies, interface errors and production incidents.

Training Outcome

Learners completing this program should be able to participate in ATTP implementation, configuration, serialization design, ECC/warehouse/packaging integration, regulatory reporting, testing and production support for pharmaceutical track-and-trace programs.